

Extracellular matrix viscoelasticity regulates spatiotemporal mammary branching morphogenesis

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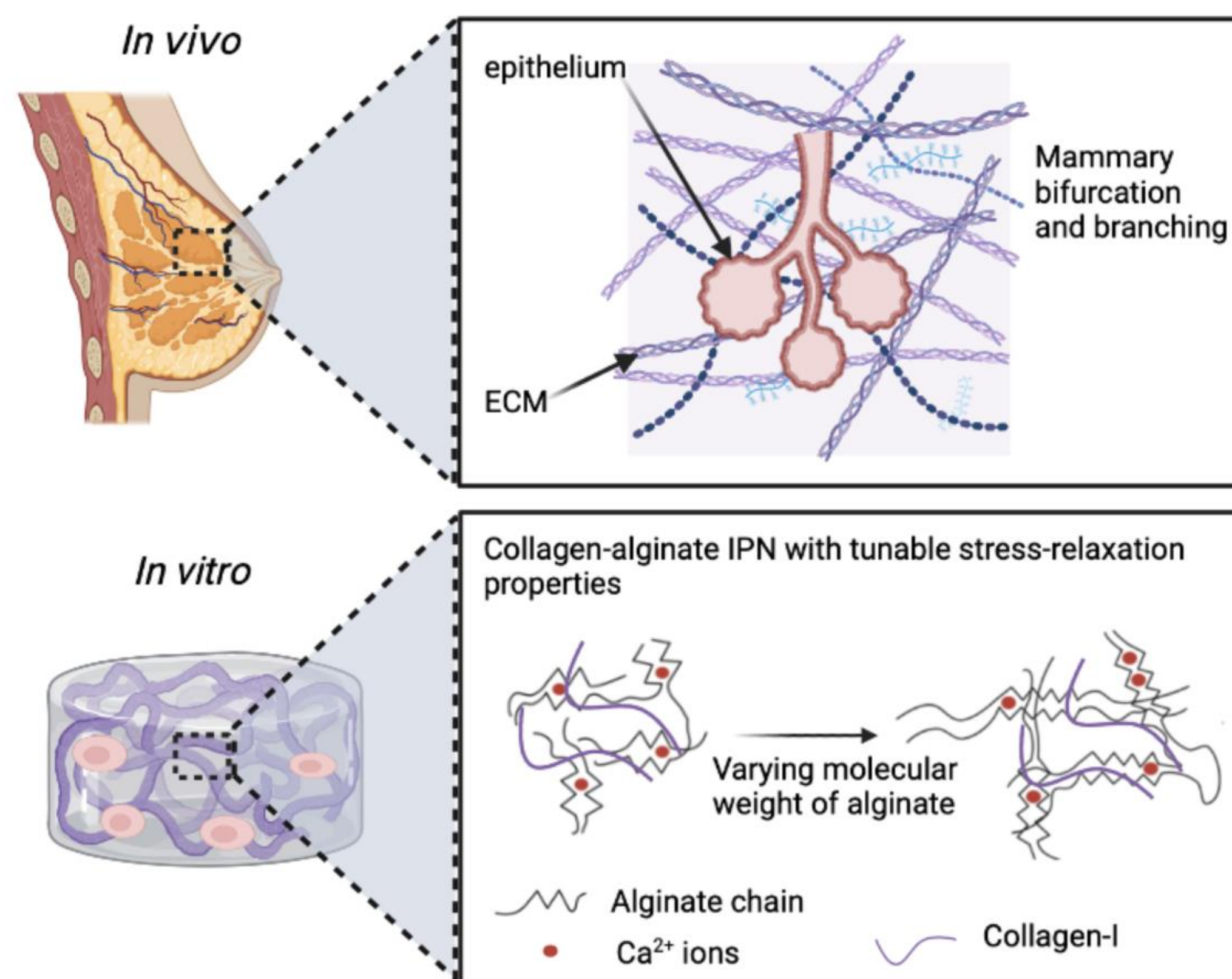
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The effect of viscoelasticity on tissue organization is unknown

Branching morphogenesis is a fundamental process that occurs during the development of mammary tissue, and several signaling pathways that regulate mammary branching are disrupted during the progression of breast cancer. Mammary gland development is influenced by mechanical and chemical cues within its *in vivo* microenvironment, and conventionally, tissue development has been studied with *in vivo* or *ex vivo/in vitro* systems where mechanical properties are poorly controlled in space and time, lack tunability, or do not mimic the native extracellular matrix (ECM) mechanics. Here, we engineered a collagen-alginate interpenetrating network (IPN) to recapitulate the mammary *in vivo* microenvironment and determine how ECM viscoelasticity regulates tissue branching in mammary development. Images created in *BioRender*.



3D collagen-alginate IPNs enable independent tuning of the viscoelasticity of the matrix

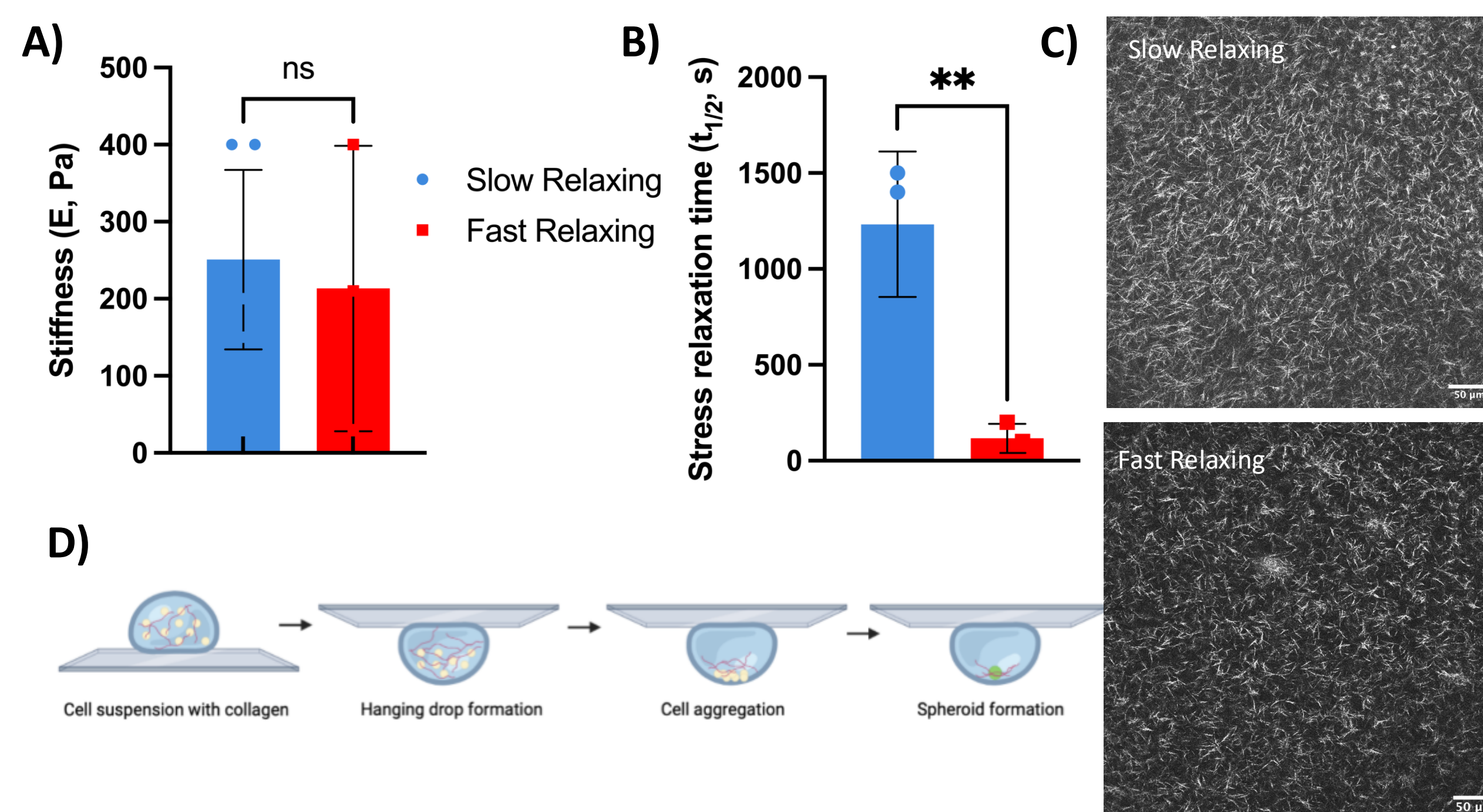


Figure 1: (A) Rheology of hydrogels reveals that slow stress-relaxing and fast stress-relaxing hydrogels have a comparable matrix stiffness that is not significantly different. (B) Molecular weight of alginate can be tuned to fabricate a slow stress-relaxing matrix and a fast stress-relaxing matrix. (C) Confocal reflectance microscopy reveals that slow relaxing matrices and fast relaxing matrices have similar collagen microarchitectures. (D) MCF10A spheroids were prepared via the hanging-drop method and encapsulated in matrices after 48 hours. Image created in *BioRender*.

Matrix mechanics influence mammary growth over 7 days

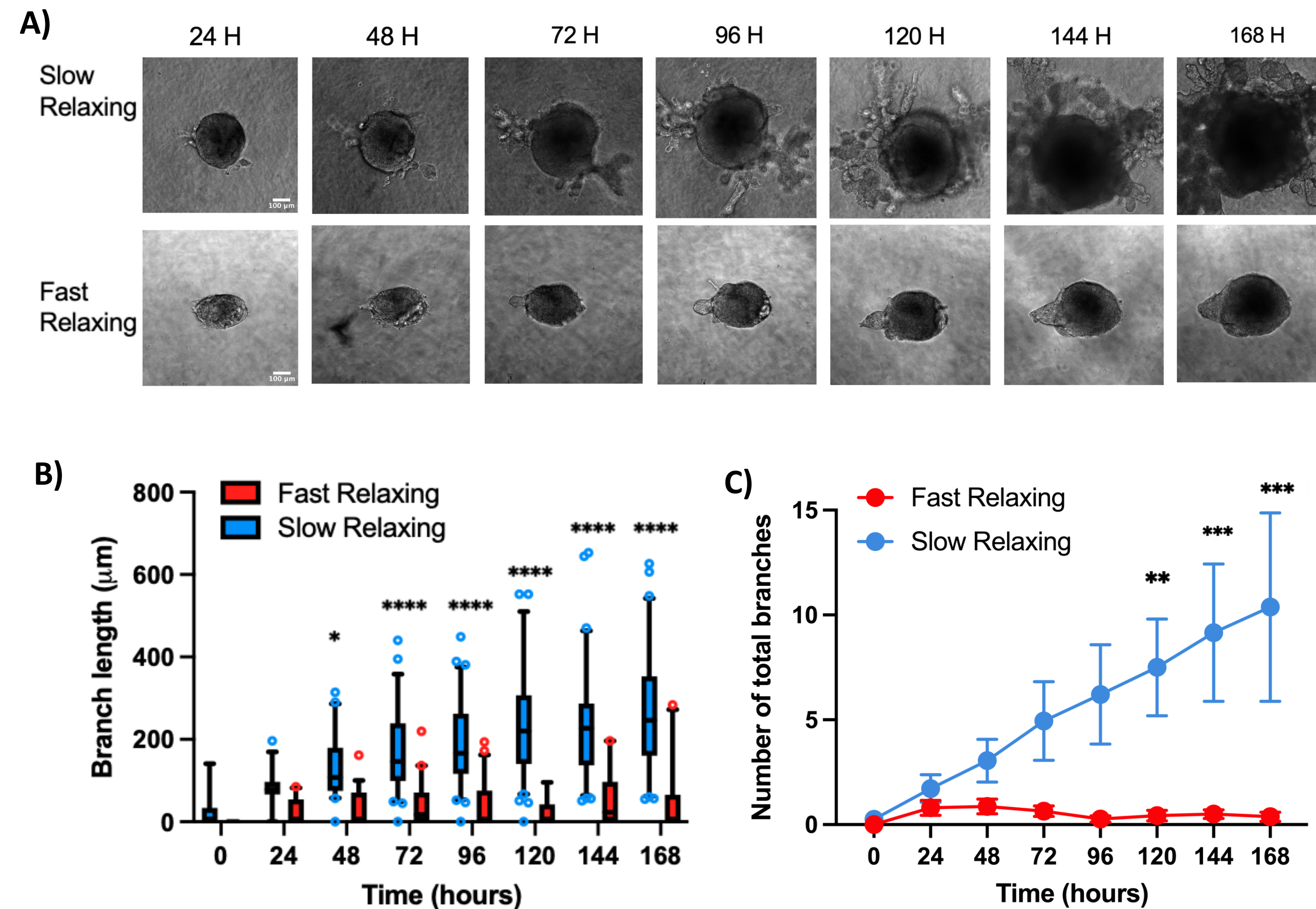


Figure 2: (A) Brightfield images of MCF10A spheroids in 3D slow stress-relaxing matrices and fast stress-relaxing matrices over 7 days. (B) Quantification of primary branch length shows a faster rate of growth in slow stress-relaxing matrices. (C) Quantification of the number of primary, secondary, and tertiary branches demonstrates more branches are formed in slow stress-relaxing matrices. Spheroid morphology and branching were analyzed and quantified using *Image J*.

Slow stress-relaxing matrices enhance spheroid area

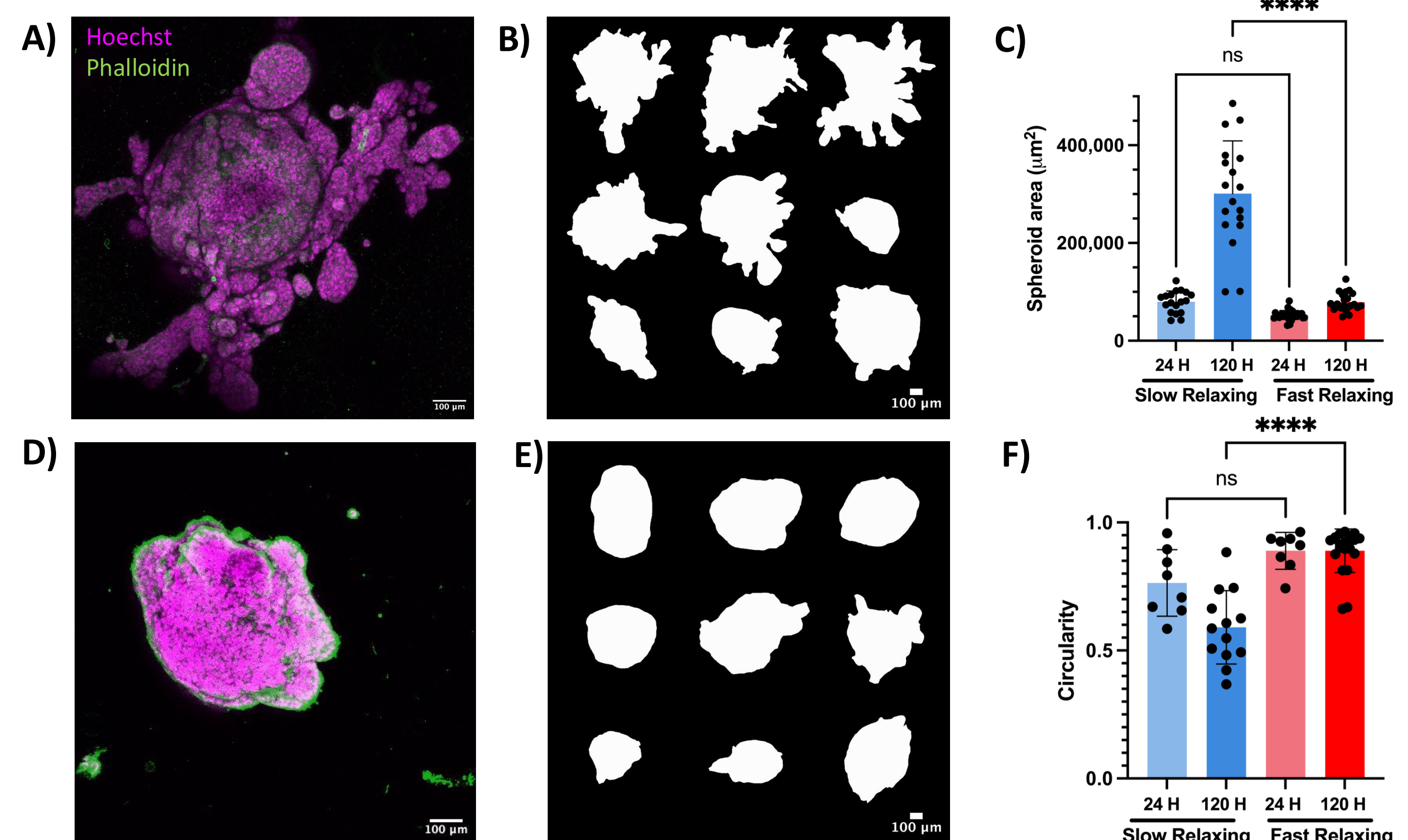


Figure 3: (A) Confocal image of a MCF10A spheroid budding and branching in a slow stress-relaxing matrix (magenta=nuclei, green=actin). (B) Representative outline of MCF10A spheroid morphologies in slow stress-relaxing matrices. (C) MCF10A spheroid area in slow stress-relaxing matrices was significantly larger after 120 hours compared to 24 hours in culture, whereas the spheroid area in fast stress-relaxing matrices between these time scales were not significantly different. (n=18 and 21, respectively). (D) Confocal image of a MCF10A spheroid in a fast stress-relaxing matrix. (E) Representative outline of MCF10A spheroid morphologies in fast stress-relaxing matrices. (F) Quantification of circularity of MCF10A spheroids. Spheroid area and circularity were quantified using *Image J*.

Slow stress-relaxing matrices promote collagen fiber alignment along longitudinal axis

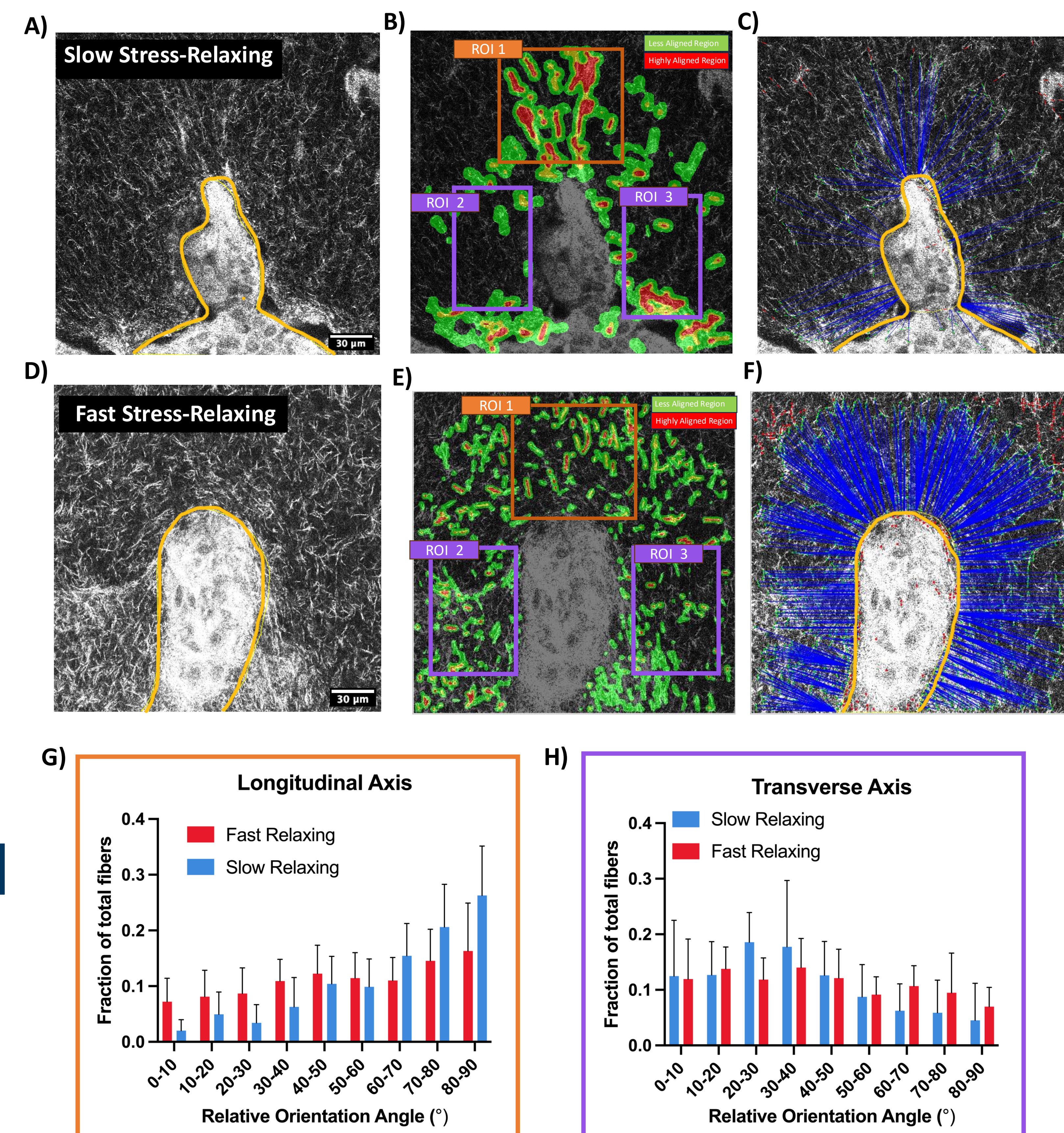


Figure 4: (A) Confocal reflectance microscopy reveals collagen fiber network surrounding a MCF10A spheroid branch in a slow stress-relaxing matrix (yellow=spheroid boundary). (B) Heatmap demonstrating regions of high and low collagen fiber alignment in a slow stress-relaxing matrix (red=high alignment, green=low alignment). ROI1 represents quantifications along the longitudinal axis and ROI 2 and 3 represent quantifications along the transverse axis. (C) Collagen fiber overlay tracks the angle between the fiber and the tangent angle of the boundary closest to fiber location in a slow stress-relaxing matrix (blue=vector for one fiber). (D-F) Representative images of collagen fiber alignment for a bud in a fast stress-relaxing matrix. (G) Quantification of relative orientation angle demonstrates that collagen fibers are highly aligned in slow-relaxing matrices along the longitudinal axis. (H) Collagen fibers display random orientation in slow and fast stress-relaxing matrices along the transverse axis. Collagen alignment was quantified via *CurveAlign*.

Conclusions and acknowledgments

- 3D alginate-collagen interpenetrating networks (IPNs) mimic *in vivo* microenvironment and enable independently tuning of the stress-relaxation of the matrix.
- Extracellular matrix viscoelasticity regulates spatiotemporal epithelial tissue organization within human mammary epithelial cells

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